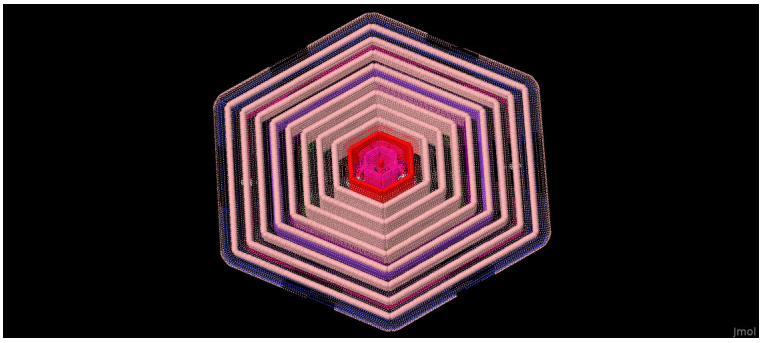
Representation and Construction of Perfects

An Illustrative Exposition of Fermatś Last Theorem and Bealś Conjecture



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Abstract

This is the only fact needed for theorem. Having two or more identical sources with which to form the perfect constrains construction to two or more layers at a time which results in the exclusion of the apex. To place the apex would require having an imbalance of points.

Author's Note

Keeping things as simple as possible, elliptic curves and cyclomatic numbers are avoided. This article does not require advanced mathematics to understand. This article does not make reference to any other work. Representation of perfects and the construction presented show solution. Both conjectures are solved by the same approach. One explanation. The conjectures are put into a form which show that the operation remove or blend on two perfects defy exact reconstruction of a perfect. The proof is simply from the definition of perfects. The mechanism is derived from a faithful representation describing all cases of the constructive model. The mechanism shown demonstrates the impossibility of construction within the constraints. The constraint of having symmetric identical sources. During construction balance is maintained. When the perfect is constructed one layer at a time, from one source, placing the apex is unconstrained. The mechanism's constraint is a property of bilateral symmetry (bls). To have bls, the formation is required to have left and right mirror symmetric sections flanking the center with an apex atop. The center is a linear arrangement of unit points. The center provides the notion of dimension. The fulcrum is the cornerstone of the constructible object. The apex is the last point placed.

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1 Nomenclature

Perfect: Δ, an object with whole number of components and a whole dimensional value which is the nth-order root of the objects measure. The 2d model is a triangle and the 3d model is a 6-sided pyramid.
Dimension: β, the notion of size.
Order: θ, the degree of the dimensional value.
Measure: v. To count the components. n. The count of components.
Point: The smallest unit comprising the components.
Apex: The unique point which completes the perfect. It is not in a layer.
Fulcrum: The first point placed; starting point upon which the Δ^x forms (grows)
Layer: The rings from edge to center (horizontal).
Ring: The hexagonal arrangement of characteristic weights. Ring 1 surrounds the fulcrum in the first layer.
Center: The 1p components which comprise the axial(C_1) or plane(C_2) center of symmetry.
Residue: The hexagonal component formed by removing $\Delta^{\theta-2}$ from Δ^θ.
Cluster: The hexagonal component formed by removing $C_{smaller}$ from Δ^θ.
Strata: The ring component index. order = 2 * strata + 1
Petal: A single layer triangular form. 6 per layer
Wedge: A multi-layered object composed of petals of diminishing size. 6 per Δ
Shell: The outer rings around center from apex to fulcrum (vertical).
Cleave: The outcome of a top difference operation forms an object with a missing top piece. This object has been 'cleaved'.
Divet: The outcome of a bottom difference operation forms an object with a missing bottom piece. This object has a 'divet'.

Perfects are not defined by length. The notion of distance is useless. Perfects do not have length, area or volume. The components of the perfect are countable arrangements of points. Counting points is to *weigh*. The measure or weight of a component is the count of points in that component. The value of the measure of the perfect is the total weight of points.

2 Problems

$$S \begin{cases} A^2 + B^2 = C^2 & \text{Pythagorean Theorem} \\ A^N + B^N = C^N & \text{Fermat's Last Theorem} \\ A^X + B^Y = C^Z & \text{Beal's Conjecture} \end{cases}$$

3 Computational Algorithm

$$H_\Delta \begin{cases} 0 & \theta \text{ zero} \\ \Delta(\theta,\beta,\delta) & \theta,\beta \text{ and } \delta \text{ whole} \end{cases}$$

$$\Delta(\theta,\beta,\delta) = (\beta+\delta)\Delta((\theta-1),\beta,\delta) + \delta\sum_{r=1}^{\beta-1}\Delta((\theta-1),r,\delta)$$

$$\begin{aligned} C_\theta &= P^\theta - P^{(\theta-2)} \\ C_\theta &= P^{(\theta-2)}(P^2-1) \\ C_\theta &= P^{(\theta-2)}(P-1)(P+1) \\ C_\theta &= P^{(\theta-3)}(P-1)(P)(P+1) \end{aligned}$$

$$C_\theta \bmod 6 \equiv 0$$

$$C_\theta \quad \text{has six identical parts when } \theta > 2$$

Calculate ring unit for single strata

sub cluster is removed to expose layers and rings

H is the weight of a single strata of a layer when $\delta = 1$

$$\begin{aligned} L0_\beta &= \Delta(\theta,\beta+0,\delta=1) - \Delta(\theta-2,\beta+0,\delta=1) \\ L1_\beta &= \Delta(\theta,\beta+1,\delta=1) - \Delta(\theta-2,\beta+1,\delta=1) \\ L2_\beta &= \Delta(\theta,\beta+2,\delta=1) - \Delta(\theta-2,\beta+2,\delta=1) \end{aligned}$$

R is the differences in adjacent layers which characterize the ring

$$\begin{aligned} R1_\beta &= (L1_\beta - L0_\beta) \\ R2_\beta &= (L2_\beta - L1_\beta) \end{aligned}$$

$$\epsilon_\psi = (R2_\beta - R1_\beta)$$

$$\epsilon_\theta = \begin{cases} 1 & \theta = 2(flat) \\ \epsilon_\psi/6 & \theta > 2 \end{cases}$$

Equations shown here define the $H_p^y e$ mechanism. See Tables 1 2 3.

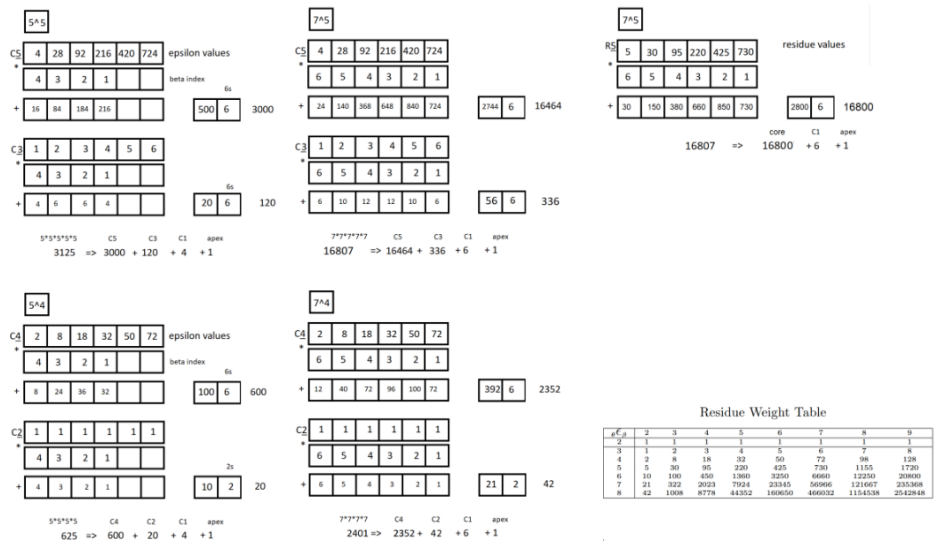

Figure 1: sub-polynomial clusters C_θ all cluster C_θ have same form; solid 3d hexagonal pyramids comprised of six wedges which form the $\Sigma\ C_n$ residue

The mechanism generates the complete set of differences for perfects of all orders. The difference of two perfects is an additive method. The recursive function uses only addition. Perfects are formed from the subpolynomial clusters, C_θ. The cluster is shown to have a 6 symmetry(6s). The epsilon weights, ϵ_ψ are 6s characteristics of a layer which define the weight of the rings in the cluster.

4 Representation and Construction

The perfect Δ^2 can be represented as a triangular arrangement of points. See Fig 4. The perfect Δ^3 can be modeled in a hexagonal pyramid arrangement of points. See Fig 5. The dimension or height of the perfect is the count of points in the center along with the apex; the ϵ_0 column. The number of layers is the count of points in the C_1 center. An even perfect Δ^E has a plane of symmetry about the C_2 center. An odd perfect Δ^O has axial symmetry around the C_1 center. See Fig 6 and Fig 8. The C_2 cluster is 2s around the C_1 center for bilateral symmetry. See Fig 7. The odd orders are transformed into the same bilateral representation. Each 6s 3d wedge is represented as two 3p ϵ units around the C_1 center.

From the diagrams it can be seen that the difference operation is two sided and produces two symmetric parts which are avialable for rearrangement into the perfect $\Delta_A = \Delta_C$ - Δ_B. See Fig 12 and Fig 13. There are three obstacles one must hurdle in order to rearrange points after the difference operation. First the 6s3p left and right core pieces are formed from 3p components, then the center must be formed from 1p units and finally an apex is placed. One stipulates to the first two of these conditions. Constructing the center requires that ϵ^θ units are decomposed into 1p units. The constraint of left and right sources force the center to be even. The center is formed from points in pairs. The newly forming object must be built two layers at a time. A 3p6s core can be formed for some measure of Δ^B and Δ^C. An even center may be placed, however infrequent. But finally to place the apex is a challenge. The apex can not be placed atop the ϵ_0 column. See Fig 9.
A 2s3p core plus even constrained center plus apex results in an odd number of components.

$$\Delta^{odd} = 1 + C_1 + C_3 + C_5 + \dots$$
$$\Delta^{even} = 1 + C_2 + C_4 + C_6 + \dots$$

$$\Delta^{odd} = apex + center + C_3 + C_5 + \dots$$
$$\Delta^{even} = apex + center + twotriangles + C_4 + C_6 + \dots$$

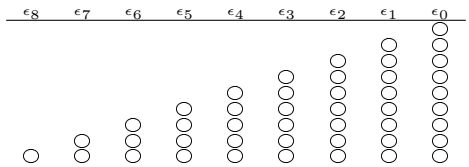

Figure 2: Axial Symmetric Basis θ odd
This figure shows a 6p ϵ_n wedge and the 1p ϵ_0 center. Six wedges and the axial center comprise the perfect.

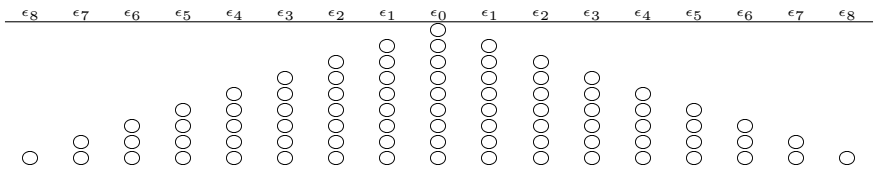

Figure 3: Plane Symmetric Basis θ even
This figure shows two 3p wedges and the 1p center. Three pairs of wedges and the planar center comprise the perfect.

5 Divet

Another difference operation, removes the apexs from both objects, ΔC and ΔB, prior to making the divet. A fulcrum is left in the divet. The missing divet is a constructable apex-less Δ_D similiar but smaller then ΔB. The divet is replenished with 3p points from the objects outer edges and/or c2 center difference for even order. Some 3p points are fractured to fill the 1p center(two layers at a time). With divet and center complete the object forms an apex-less $\Delta_C - \Delta_B = \Delta_A - 1$. See Fig 12

ψ	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1
2	1	3	5	7	9	11	13	15	17
3	1	7	19	37	61	91	127	169	217
4	1	15	65	175	369	671	1105	1695	2465
5	1	31	211	781	2101	4651	9031	15961	26281
6	1	63	665	3367	11529	31031	70993	144495	269297
7	1	127	2059	14197	61741	201811	543607	1273609	2685817
8	1	255	6305	58975	325089	1288991	4085185	11012415	26269505

Table 1: Layer Weights
the difference of two perfects $\Delta^{\psi}_{N+1} - \Delta^{\psi}_{N}$ forms when removed at top a layer or when removed at bottom a shell.

ϵ_ψ	2	3	4	5	6	7	8	9
2	1	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	4	28	92	216	420	724	1148	1712
6	8	92	432	1328	3200	6588	12152	20672
7	16	292	1928	7704	22920	56236	120512	233648
8	32	908	8328	42992	157400	459372	1142288	2522048

Table 2: ϵ weights : single strata weight units for rings

ϵ_ψ	2	3	4	5	6	7	8	9
2	1	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	5	30	95	220	425	730	1155	1720
6	10	100	450	1360	3250	6660	12250	20800
7	21	322	2023	7924	23345	56966	121667	235368
8	42	1008	8778	44352	160650	466032	1154538	2542848

Table 3: Residue Weight Table
sum of epsilon weights for all strata that form cluster

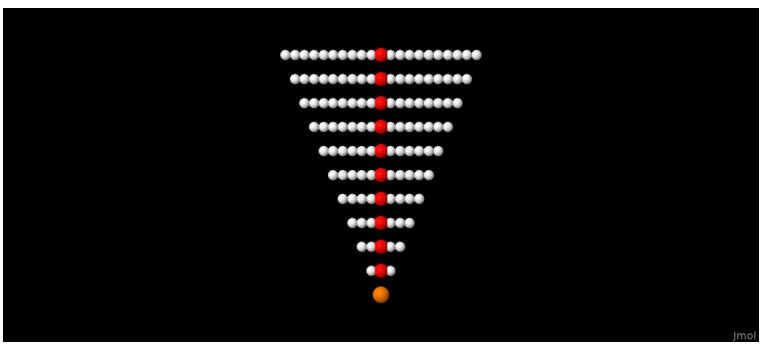

Figure 4: Perfect Δ^2 arranged as a triangular square. Each point has a unit weight for a total weight of $N^2 = 1 + 3 + 5 + \ldots + (2N - 1)$.

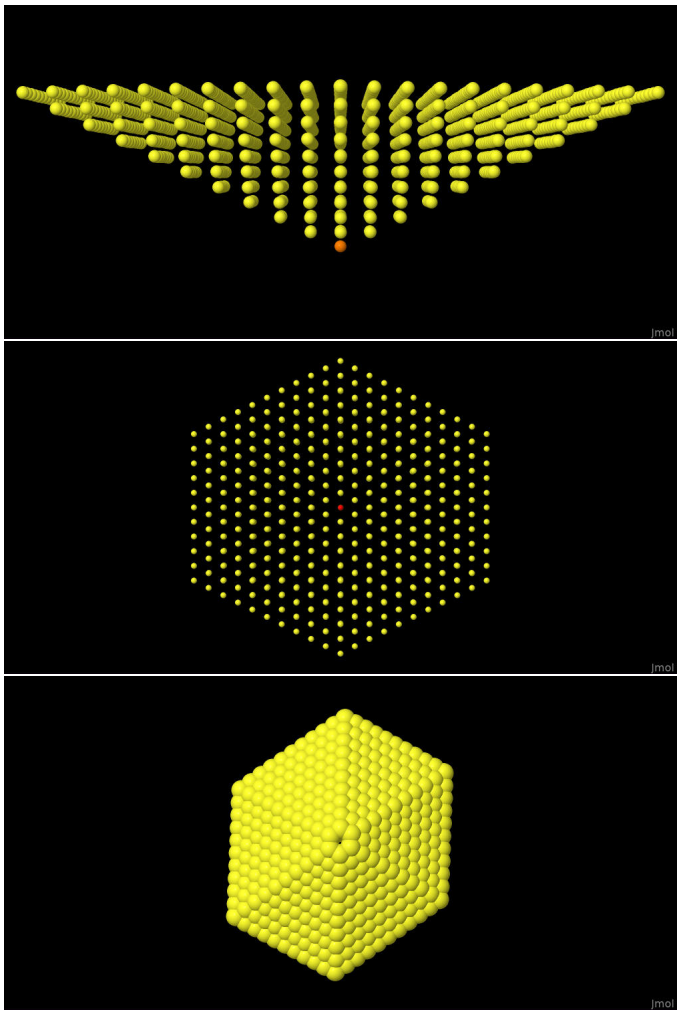

Figure 5: Perfect Δ^3 arranged as a hexagonal pyramid. Each point has a unit weight for a total weight of $N^3 = 1 + 7 + 19 + \ldots$.

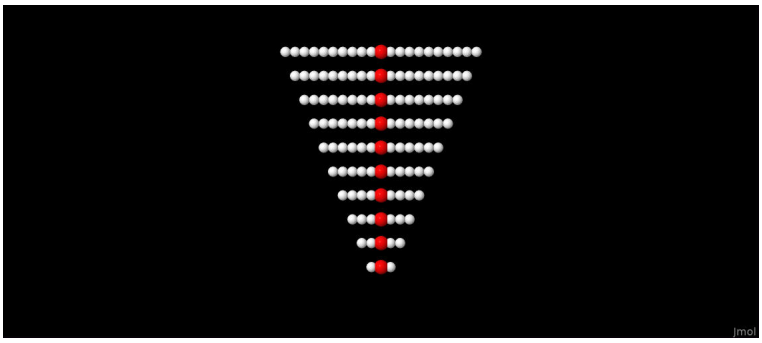

Figure 6: c2 center is the plane of symmetry in even order Δ

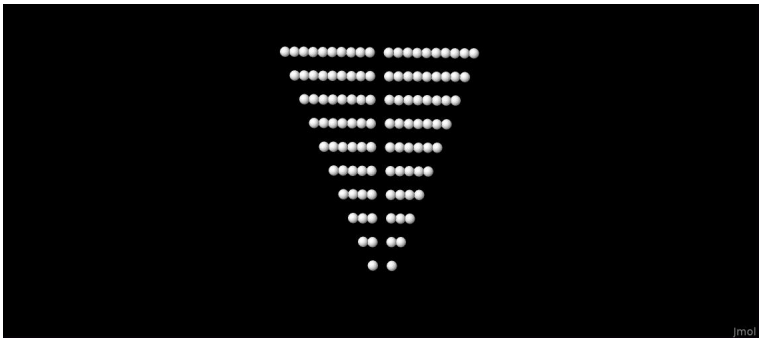

Figure 7: Removal of c1 center exposes two symmetric triangles which are identical. These places have weight and measure as one point. C2 is only present in even orders. C1 is present in all orders. Using this breakage, perfects are represented in a flat bi-lateral form which allows for a single case to examine.

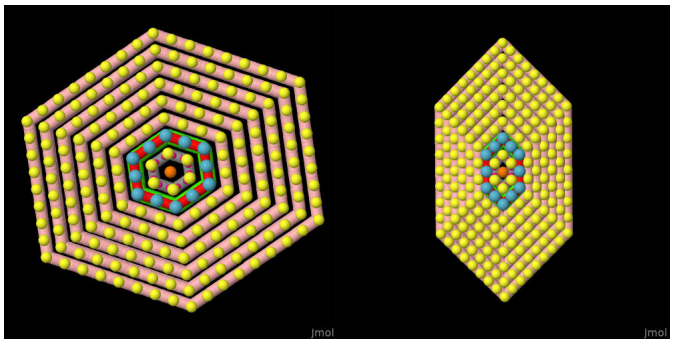

Figure 8: axial and planar symmetry

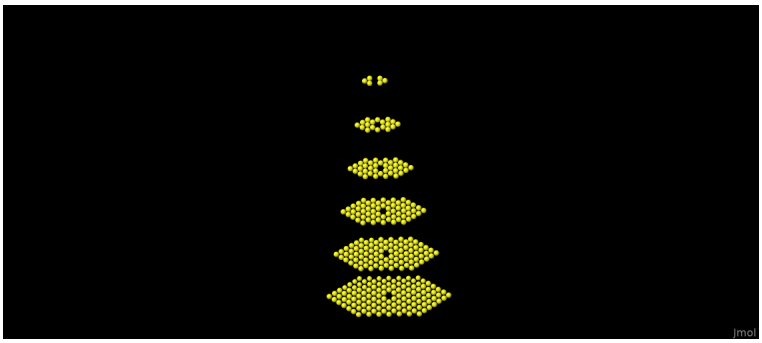

Figure 9: $P^3 - P^1 = C_3$

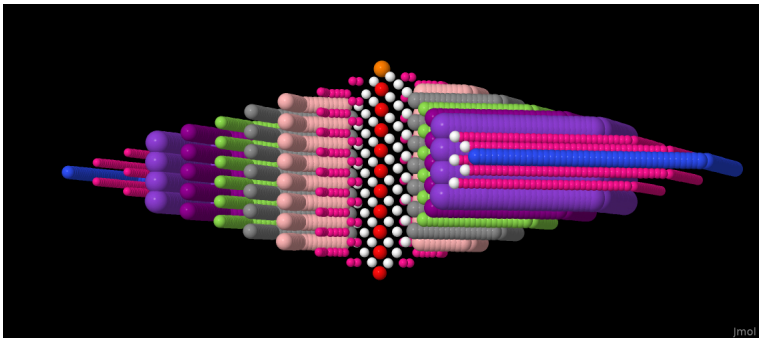

Figure 10: squarehex

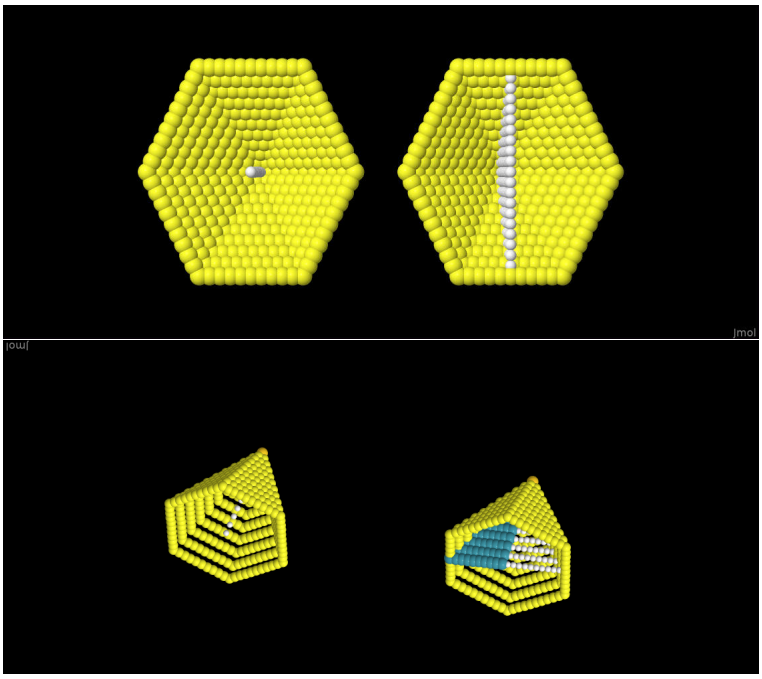

Figure 11: interhex

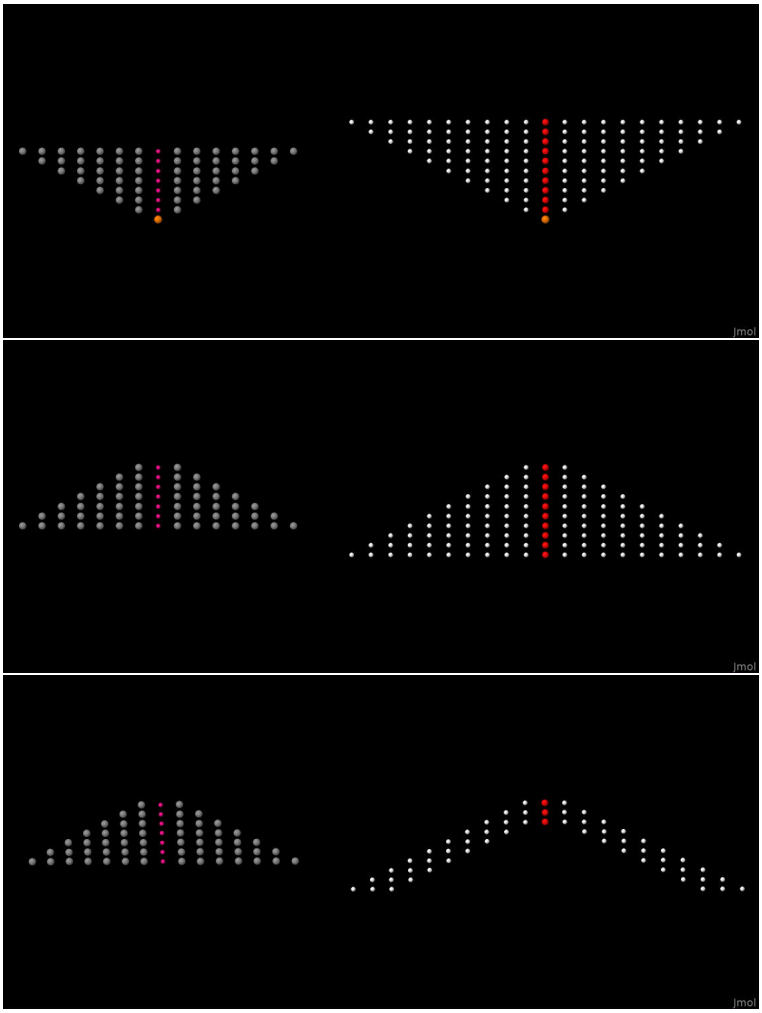

Figure 12: c2delhex

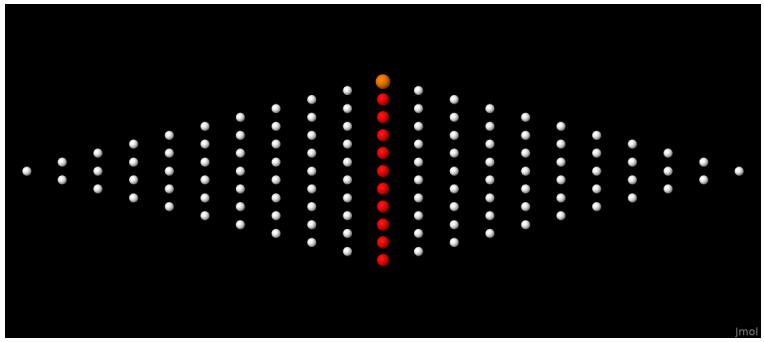

Figure 13: c2difhex $\epsilon 0 * \epsilon 1 * \epsilon 2$

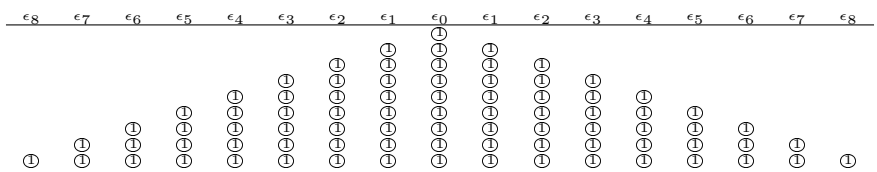

Figure 14: Δ^2 flat representation

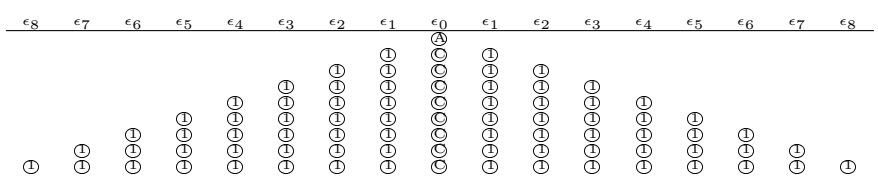

Figure 15: Δ^θ flat bilateral representation showing apex and center
θ odd w(A)=1 w(C)=1 $\Sigma w(1) = \Sigma(\epsilon_p^\theta)$
θ even w(A)=1 w(C)=1 $\Sigma w(1) = \Sigma(\epsilon_p^\theta) + 1$
one for each place in the two triangles of the broken form.